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(54) Title: PHARMACEUTICAL COMPOSITIONS OF ACOTIAMIDE AND PROTON PUMP INHIBITOR

(57) Abstract: The invention relates to pharmaceutical unit compositions comprising multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salt thereof and a preparation method thereof. The invention further relates to pharmaceutical unit compositions comprising modified release multi-particulate compositions of acotiamide or pharmaceutically acceptable salts thereof and delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof. Methods of preparing such compositions are also provided. The compositions can be administered once daily.

WO 2020/104955 A1

PHARMACEUTICAL COMPOSITIONS OF ACOTIAMIDE AND PROTON PUMP INHIBITOR

TECHNICAL FIELD OF THE INVENTION

The present specification relates to pharmaceutical unit compositions comprising multi-particulate modified release compositions of acotiamide. The specification also relates to pharmaceutical unit compositions comprising acotiamide and a proton pump inhibitor. Methods of preparing such compositions are also provided.

BACKGROUND OF THE INVENTION

Gastroesophageal reflux disease (GERD) is defined as the reflux of gastric contents into the esophagus causes symptoms severe enough to adversely affect quality of life. Functional dyspepsia (FD) is distinct from GERD is defined as conditions in which upper abdominal symptoms are present in the absence of any organic disease that may explain the symptoms. According to the Rome III criteria, the international diagnostic criteria, FD is defined as "a disease that started at least 6 months prior to diagnosis, showing at least one of the following symptoms for the last 3 months: bothersome postprandial fullness, early satiation, epigastric pain, epigastric burning; and there is no evidence of structural disease (including at upper endoscopy) that is likely to explain the symptoms."

The cause of FD is not fully elucidated. The current assumption is that the symptoms are caused by the involvement of multiple factors including abnormal gastric emptying, gastric dysrhythmia, gastric hypersensitivity, hypersensitivity and motility disorder of the small intestine, impaired postprandial relaxation of the gastric fundus, vagus nerve disorder, enhanced acid sensitivity, psychological factors, and central nerve disorders.

Acotiamide is a first-in-class drug that is used to treat functional dyspepsia (FD). Acotiamide is an acetylcholinesterase (AChE) inhibitor discovered by the Zeria Pharm. Acotiamide suppresses the degradation of acetylcholine (ACh) released from cholinergic nerve terminals by inhibiting AChE, thereby enhancing the ACh-induced contraction and motility of the gastric antrum and the gastric body. It is considered that acotiamide acts as an antagonist on muscarinic autoreceptors in the enteric nervous system and inhibits acetylcholinesterase activity.

US patent 5,981,557 discloses aminothiazole derivative, medicament containing the same, and intermediate for preparation of said compound. Also discloses method for the prevention and treatment of diseases caused by digestive dysmotility.

Acotiamide hydrochloride hydrate is approved in Japan as 100 mg oral tablet and marketed under brand name ACOFIDE® for the treatment of postprandial fullness, upper abdominal bloating, and early satiation due to functional dyspepsia. A favorable clinical course with acotiamide 100 mg three times daily was demonstrated with high symptom elimination rate for patients of FD.

The usual daily recommended adult dosage of ACOFIDE® is 300 mg, which is achieved by administering 100 mg tablet three times daily before a meal. It is given three times a day to maintain the plasma concentration. Frequent administration is a major drawback for acotiamide treatment. Currently there is no approved single 300 mg unit composition of acotiamide which could be administered once daily. Thus, there remains an unmet need for once daily single unit acotiamide composition which improves patient compliance and convenience on taking of 300 mg once instead of three times a day and maintain the plasma concentration as well.

Proton pump inhibitors (PPIs) are among the most widely prescribed and over-the-counter for heartburn, gastroesophageal reflux disease (GERD) and other upper gastrointestinal issues. Examples of proton pump inhibitors are include but not limited to lansoprazole, omeprazole, esomeprazole, rabeprazole, pantoprazole,

leminoprazole, or an optically active isomer thereof, or a pharmaceutically acceptable salt thereof.

PPIs are often the first choice for treating FD. However, some patients need additional medication because of residual symptoms despite a certain level of benefit from the PPIs. For these patients, a combination of PPI and other agents has a possibly more beneficial effect than changing their medication. Currently, there is no approved combination drug product available for acotiamide and a proton pump inhibitor in a single composition. Thus, there remains an unmet need for once daily oral composition comprising combination of acotiamide and a PPI in a single composition, wherein the said composition may be effective for the management and treatment of FD, GERD, and other upper gastrointestinal issues or disorders.

IN 201721028736 disclosed direct compressible dual drug releasing pharmaceutical composition comprising extended release acotiamide and immediate release proton pump inhibitors. The microcapsule of acotiamide are compressed along with bare proton pump inhibitor to form a final tablet.

It is often difficult to prepare single unit composition of a suitable swallow able size when the composition is having relatively high amount of drug. Therefore a high drug load compositions are being preferred for such cases. However, compositions containing high drug load are challenging to prepare, especially when they have a modified release profile. Therefore, there remains a need for improved pharmaceutical composition of acotiamide with improved patient compliance, reduced frequency of administration and should not to be too large to avoid swallowing difficulties.

Accordingly, the present invention provides once daily multi-particulate acotiamide compositions. The present inventors also surprisingly found that the modified release composition for once daily administration can also be combined with a delayed release proton pump inhibitor in the single unit composition. This represents a substantial improvement in patient compliance. Accordingly, the present invention

also provides once daily pharmaceutical unit compositions comprising multi-particulate compositions of acotiamide and a proton pump inhibitor combination.

OBJECT OF THE INVENTION

The present specification relates to multi-particulate pharmaceutical compositions of acotiamide.

In one aspect, the present specification relates to multi-particulate modified release compositions of acotiamide.

In another aspect, the present specification relates to multi-particulate sustained release and/or delayed release compositions of acotiamide.

In yet another aspect, the present specification relates to multi-particulate immediate release and sustained release and/or delayed release compositions of acotiamide.

In yet another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof, and one or more rate controlling polymers,
ii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily and the total dose of acotiamide or pharmaceutically acceptable salts thereof in the unit composition is 300 mg.

In yet another aspect, the present specification relates to multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof, wherein acotiamide or pharmaceutically acceptable salts thereof is present in an amount of at least 45%, preferably at least 50%, more preferably at least 60% by weight of the total unit composition.

In yet another aspect the present specification relates to multi-particulate modified release compositions of acotiamide when subjected to *in vitro* changeover dissolution study exhibits following release profile:

i) releases not more than 55% in 1 hour, in 900 ml of 0.01N HCl medium at 100 rpm and 37°C using USP type I apparatus, and
ii) releases at least about 20% - 75% in 3 hours and not less than 50% in 9 hours, in 900 ml of pH 6.8 phosphate buffer medium saline with 0.5% Sodium laureth sulfate at 100 rpm and 37°C using USP type I apparatus.

In one aspect, the present specification relates to pharmaceutical unit composition comprising acotiamide and a proton pump inhibitor.

In another aspect, the present specification relates to once daily pharmaceutical unit composition comprising acotiamide and a proton pump inhibitor.

In one aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) plurality of multi-particulate immediate release and/or modified release compositions of acotiamide and one or more rate controlling polymer, and
ii) delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily and the total dose of acotiamide or a pharmaceutically acceptable salts thereof in the unit composition is 300 mg.

In another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or a pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the acotiamide or a pharmaceutically acceptable salts thereof is present in an amount of at least 50% by weight of the total multi-particulate acotiamide compositions.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide and one or more rate controlling polymer, and
ii) delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate immediate release and sustained release and/or delayed release compositions of acotiamide, and
ii) delayed release compositions of a proton pump inhibitor.

In yet another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide, wherein total dose of said compositions is 300 mg acotiamide and
ii) delayed release compositions of a proton pump inhibitor,

wherein the said capsule unit composition is administered once daily.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts,

wherein the unit composition when subjected to *in vitro* changeover dissolution study exhibits the following release profile of acotiamide or pharmaceutically acceptable salts thereof:

i) releases not more than 55% in 1 hour, in 900 ml of 0.01N HCl medium at 100 rpm and 37°C using USP type I apparatus, and
ii) releases at least about 20% - 75% in 3 hours and not less than 50% in 9 hours, in 900 ml of pH 6.8 phosphate buffer medium saline with 0.5% Sodium laureth sulfate at 100 rpm and 37°C using USP type I apparatus.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) one or more sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) multi-particulate delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) four sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) one delayed release mini-tablet of a proton pump inhibitor.

In another aspect, the present invention relates to a capsule unit composition comprising:

i) one or more immediate release mini-tablets of acotiamide and one or more sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) multi-particulate delayed release compositions of a proton pump inhibitor.

The present specification also relates to use of multi-particulate pharmaceutical compositions comprising acotiamide and a proton pump inhibitor for the treatment and/or management of functional dyspepsia, postprandial fullness in the functional

dyspepsia, upper abdominal fullness, gastroesophageal reflux disease, and other upper gastrointestinal issues or disorders.

BRIEF DESCRIPTION OF FIGURES

Figure 1: Dissolution profile of pharmaceutical unit compositions comprising multi-particulate modified release Acotiamide HCl hydrate at initial stage.

DESCRIPTION OF INVENTION

The present specification relates to multi-particulate pharmaceutical compositions of acotiamide.

In one aspect, the present specification relates to multi-particulate modified release compositions of acotiamide.

In another aspect, the present specification relates to multi-particulate sustained release and/or delayed release compositions of acotiamide.

In yet another aspect, the present specification relates to multi-particulate immediate release and sustained release and/or delayed release compositions of acotiamide.

In yet another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
ii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily and the total dose of acotiamide or pharmaceutically acceptable salts thereof in the unit composition is 300 mg.

In yet another aspect, the present specification relates to multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof, wherein acotiamide or pharmaceutically acceptable salts thereof is present in an amount of at least 45%, preferably at least 50%, more preferably at least 60% by weight of the total unit composition.

In yet another aspect the present specification relates to multi-particulate modified release compositions of acotiamide when subjected to *in vitro* changeover dissolution study exhibits following release profile:

ii) releases not more than 55% in 1 hour, in 900 ml of 0.01N HCl medium at 100 rpm and 37°C using USP type I apparatus, and
ii) releases at least about 20% - 75% in 3 hours and not less than 50% in 9 hours, in 900 ml of pH 6.8 phosphate buffer medium saline with 0.5% Sodium laureth sulfate at 100 rpm and 37°C using USP type I apparatus.

In one aspect, the present specification relates to pharmaceutical unit composition comprising acotiamide and a proton pump inhibitor.

In another aspect, the present specification relates to once daily pharmaceutical unit composition comprising acotiamide and a proton pump inhibitor.

In one aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) plurality of multi-particulate immediate release and/or modified release compositions of acotiamide and one or more rate controlling polymer, and
ii) delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily and the total dose of acotiamide or pharmaceutically acceptable salts thereof in the unit composition is 300 mg.

In another aspect, the present specification relates to a pharmaceutical unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts,

wherein the acotiamide or pharmaceutically acceptable salts thereof is present in an amount of at least 50% by weight of the total multi-particulate acotiamide compositions.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide and one or more rate controlling polymer, and
ii) delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate immediate release and sustained release and/or delayed release compositions of acotiamide, and
ii) delayed release compositions of a proton pump inhibitor.

In yet another aspect, the present specification relates to a capsule unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide, wherein total dose of said compositions is 300 mg acotiamide and
ii) delayed release compositions of a proton pump inhibitor,

wherein the said capsule unit composition is administered once daily.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,
ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
iii) optionally multi-particulate immediate release compositions of acotiamide or a pharmaceutically acceptable salts,

wherein the unit composition when subjected to *in vitro* changeover dissolution study exhibits the following release profile of acotiamide or pharmaceutically acceptable salts thereof:

ii) releases not more than 55% in 1 hour, in 900 ml of 0.01N HCl medium at 100 rpm and 37°C using USP type I apparatus, and

ii) releases at least about 20% - 75% in 3 hours and not less than 50% in 9 hours, in 900 ml of pH 6.8 phosphate buffer medium saline with 0.5% Sodium laureth sulfate at 100 rpm and 37°C using USP type I apparatus.

. In another aspect, the present specification relates to a capsule unit composition comprising:

i) one or more sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and

ii) multi-particulate delayed release compositions of a proton pump inhibitor.

In another aspect, the present specification relates to a capsule unit composition comprising:

i) four sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and

ii) one delayed release mini-tablet of a proton pump inhibitor.

In another aspect, the present invention relates to a capsule unit composition comprising:

i) one or more immediate release mini-tablets of acotiamide and one or more sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and

ii) multi-particulate delayed release compositions of a proton pump inhibitor.

The term “pharmaceutical composition” as used herein refers to oral dosage forms preferably in the form of tablets, capsules, sachets and the like. The dosage forms generally are prepared by using suitable pharmaceutically acceptable excipients. The unit composition refers to a single unit of such pharmaceutical compositions.

The term “acotiamide” as used in the context of the present specification relates to the free base form, acid form, salt form, polymorphic crystalline or amorphous form, solvates, hydrates, ethers, esters, etc. Preferably acotiamide is present in the salt form, e.g. acotiamide hydrochloride or acotiamide hydrochloride hydrate. The therapeutic effective dose of acotiamide or pharmaceutically acceptable salts thereof is 50mg, 100mg, 150mg, 200mg, 250mg, 300mg, 350mg and 400mg. A standard dose of 300 mg per day has been proven clinically effective for the treatment of functional dyspepsia.

The term “multi-particulate” or “multi-particulate compositions” can be, but not limited to mini-tablets, spheroids, pellets, granules, powders, microcapsules, multiple unit particles, and the like. Multi-particulate can be filled in but not limited to sachet, capsules and the like or can be further processed in the form of solid dosage forms. Solid dosage forms can be but not limited to tablets, bilayer tablets, inlay tablets, capsules and the like. The preferred “multi-particulate” compositions is mini-tablets or pellets. The term “mini-tablets” as used herein refers to small round or cylindrical tablets, typically 2 to 7 mm in diameter and are produced by tableting technology, typically rotary presses with minor modifications. The mini-tablets offer finished dosage form flexibility in that they can be delivered as capsules, sachets or compressed into larger tablets. The size/dimension of the mini-tablets are selected in such a way, that it can be accommodated in a capsule of a suitable size.

The term “immediate release” as used herein refers to immediate release of drug from the compositions after administration. The “modified release” relates release of drug from the compositions wherein said release only occurs some times after the administration or for a prolonged period of time or to a specific target in the body. Modified release systems can be further classified as: delayed release, where drug is released only at some point after the initial administration; extended release or sustained release:-prolongs the release at a controlled rate to reduce dosing frequency. These terms are also used by the pharmacopoeias and the FDA. Whilst immediate-

release dosage forms are designed to give a fast onset of drug action, modifications in drug release are often desirable to increase the stability, safety and efficacy of the drug, to improve the therapeutic outcome of the drug treatment and/or to increase patient compliance and convenience of administration.

The term "once daily" as used herein refers to once in a day administration of the compositions to the subject in the need thereof. The once daily administration could be any time of the day before or after meal. The preferred mode of administration would be once a day before meal.

The term "proton pump inhibitors (PPIs)" as used herein refers to the compound that inhibits the gastric acid secretion by blocking the gastric H,K-ATPase (proton pump). Examples of proton pump inhibitors are include but not limited to lansoprazole, omeprazole, esomeprazole, rabeprazole, pantoprazole, leminoprazole, or an optically active isomer thereof, or a pharmaceutically acceptable salt thereof.

The term "changeover dissolution study" as used herein refers to the dissolution study which is performed by using USP Type I (Basket) apparatus with 100 RPM, and the dissolution methodology includes one acid stage followed by one buffer stage. Firstly *in vitro* dissolution was carried out in acid stage (900 ml of 0.01N HCl solution, USP Type 1 apparatus at a speed of 100 rpm and 37°C) till 60 minutes, followed by buffer stage (900 ml of phosphate buffer solution having pH 6.8 with 05% sodium laureth sodium at a speed of 100 rpm and 37°C). Multi-particulate modified release acotiamide compositions of the present specification for once daily oral administration when subjected to a changeover dissolution study releases not more than 55% in 1hour in acid stage, at least about 20% - 75% in 3 hours, and not less than 50% in 9 hours in buffer stage.

The term "pharmaceutically acceptable excipients" include, but not limited to, one or more of rate controlling polymers, diluents, disintegrants, binders, lubricants,

glidants, acidifying agent, alkalizing agent, stabilizers, surfactants, sweetener, film coating materials, plasticizers, pigments, opacifiers, coloring agents and the like.

Suitable "rate controlling polymers" may include but not limited to one or more hydrophilic polymers, hydrophobic polymers, Natural polymers, bioadhesive polymer pH-dependent or pH-independent, enteric, degradable, non-degradable, enteric polymers, melt extrusion polymers and the like.

Suitable hydrophilic polymers may include one or more of cellulosic polymers/copolymers or its derivatives including, but not limited to methyl cellulose, hydroxypropyl methylcellulose (not limited to HPMC K4M, K-100, K-200), hydroxyethyl cellulose, hydroxypropyl cellulose, hydroxyethyl methylcellulose, carboxymethylcellulose, sodium carboxymethylcellulose; polyacrylates, methyl acrylates, polyethylene oxides, polyethylene glycols, chitosan, gums, starch derivatives, polyurethanes, galactomannans, polysaccharides, polyalcohols, acrylic acid or acrylamide derivatives, and the like.

Suitable hydrophobic polymers include one or more of ethyl cellulose, glycerol palmitostearate, beeswax, glycowax, carnaubawax, hydrogenated vegetable oil, glycerol monostearate, stearylalcohol, glyceryl behenate, polyanhydrides, methyl acrylates and the like.

Natural polymers include but are not limited to proteins (e.g., hydrophilic proteins), such as pectin, zein, modified zein, casein, gelatin, gluten, serum albumin, or collagen, chitosan, oligosaccharides and polysaccharides such as cellulose, dextrans, tamarind seed polysaccharide, gellan, carrageenan, xanthan gum, gum Arabic; hyaluronic acid, polyhyaluronic acid, alginic acid, sodium alginate and the like.

Suitable bioadhesive polymers selected from but are not limited to polyamides, polycarbonates, polyalkylenes, polyalkylene glycols, polyalkylene oxides, polyalkylene terephthalates, polyvinyl alcohols, polyvinyl ethers, polyvinyl esters,

polyvinyl halides, polyvinylpyrrolidone, polyglycolides, polysiloxanes, polyurethanes, polystyrene, polymers of acrylic and methacrylic esters, polylactides, poly(butyric acid), poly(valeric acid), poly(lactide-co-glycolide), polyanhydrides, polyorthoesters, poly(fumaric acid), poly(maleic acid), and blends and copolymers or mixtures thereof and the like.

Other rate controlling polymers suitable for use in the invention include, but are not limited to, cellulose acetate, cellulose propionate, cellulose acetate butyrate, cellulose acetate phthalate, carboxymethyl cellulose, cellulose triacetate, cellulose sulfate sodium salt, poly(methyl methacrylate), poly(ethyl methacrylate), poly(butyl methacrylate), poly(isobutyl methacrylate), poly(hexyl methacrylate), poly(isodecyl methacrylate), poly(lauryl methacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate) polyethylene, polypropylene, poly(ethylene glycol), poly(ethylene oxide), poly (ethylene terephthalate), polyvinyl acetate), polyvinyl chloride, polystyrene, polyvinyl pyrrol idone, polyvinylphenol, Polylactides, polyglycolides and copolymers thereof, poly(ethylene terephthalate), poly(butyric acid), poly(valeric acid), poly(lactide-co-caprolactone), poly[lactide-co- glycolide], polyanhydrides (e.g., poly(adipic anhydride)), polyorthoesters, blends and copolymers thereof and the like. Methacrylates can be but not limited to Eudragit® L; Eudragit® S; Eudragit® FS 30 D; Eudragit® L30D-55; and Eudragit® L100-55, Eudragit RL PO, Eudragit RL 100, Eudragit RL 30 D, Eudragit ® E, Eudragit ® NE and the like.

Suitable rate controlling polymer may include one or more enteric coatings are usually formulated with synthetic polymers that contain ionisable functional groups that render the polymer water soluble at a pH value.

All enteric polymers that remain intact at pH value lower than about 4.0 and dissolve at pH values higher than 4.0, preferably higher than 5.0, most preferably about 6.0, are considered useful as rate controlling agents for this composition.

Suitable enteric polymers may include but not limited to one or more of Methyl acrylate-methacrylic acid copolymers, Cellulose acetate phthalate (CAP), Cellulose acetate succinate, Hydroxypropyl methyl cellulose phthalate, Hydroxypropyl methyl cellulose acetate succinate (hypromellose acetate succinate), Polyvinyl acetate phthalate (PVAP), Methyl methacrylate-methacrylic acid copolymers, Shellac, Cellulose acetate trimellitate, Sodium alginate, Zein, Hydroxyethyl ethyl cellulose phthalate, Cellulose acetate tetrahydrophtalate, Acrylic resin and the like.

Suitable melt extrusion polymers can be, but not limited to derivatised cellulose, poly(methacrylate) derivative, poly(ethylene-co-vinyl acetate), poly(ethylene), poly(vinyl acetate-co-methacrylic acid), epoxy resins and caprolactones, poly(ethylene oxide), poly(ethylene glycol) and others including various waxes, fats, lipid-based excipients, including the Gelucire®, Witepsol®, Labrafil® and the like.

In another aspect, the present invention relates to multi-particulate modified release composition of acotiamide and/or proton pump inhibitor one or more rate controlling polymers.

Suitable diluents, may be selected from, but not limited to the group consisting of different grades of starches, such as maize starch, potato starch, rice starch, wheat starch, pregelatinised starch, fully pregelatinised starch; cellulose derivatives, such as microcrystalline cellulose or silicified microcrystalline cellulose; sugar alcohols such as mannitol, erythritol, sorbitol, xylitol; monosaccharides like glucose; oligosaccharides like sucrose and lactose such as lactose monohydrate, lactose anhydrous, spray dried lactose or anhydrous lactose; calcium salts, such as calcium hydrogen phosphate; particularly preferably the fillers are selected from the group consisting of, microcrystalline cellulose, silicified microcrystalline cellulose, lactose monohydrate, spray dried lactose, and anhydrous lactose and the like.

Suitable disintegrants may be selected from, but not limited to the group consisting of carmellose calcium, carboxymethylstarch sodium, croscarmellose sodium (cellulose carboxymethylether sodium salt, crosslinked), starch, modified starch such as pregelatinized starch, starch derivatives such as sodium starch glycolate, crosslinked polyvinylpyrrolidone (crospovidone), and low-substituted hydroxypropylcellulose, and disintegrating aids such as magnesium alumino-metasilicate and ion exchange resins like polacrilin potassium; particularly preferably the disintegrants are selected from the group consisting of sodium starch glycolate, croscarmellose sodium and crospovidone and the like.

Suitable binders may be selected from, but not limited to the group consisting of polyvinyl pyrrolidone (Povidone), polyvinyl alcohol, copolymers of vinylpyrrolidone with other vinyl derivatives (Copovidone), hydroxypropyl methylcellulose, methylcellulose, hydroxypropylcellulose, powdered acacia, gelatin, guar gum, carbomer such as carbopol, polymethacrylates and pregelatinized starch and the like.

Suitable lubricants may be selected from, but not limited to the group consisting of stearic acid, talc, glyceryl behenate, sodium stearyl fumarate and magnesium stearate; particularly preferably the lubricant is magnesium stearate and sodium stearyl fumarate and the like.

Suitable glidants, may be selected from, but not limited to the group consisting of colloidal silica, hydrophobic colloidal silica and magnesium trisilicate, such as talc and the like.

Composition may contain acidifying and alkalinizing agent. Acidifying agents can be selected from, but not limited to citric acid, succinic acid, fumaric acid, malic acid, maleic acid, glutaric acid, lactic acid and mixtures thereof and the like. Alkalizing agent can be selected from, but not limited to magnesium oxide, aluminium oxide, ammonium hydroxide, magaldrate, an alkali metal salt or alkaline earth metal salt, such

as sodium bicarbonate, calcium carbonate or sodium citrate, an alkali metal hydroxide such as sodium hydroxide, potassium hydroxide or lithium hydroxide, or an alkaline earth metal hydroxide such as calcium hydroxide or magnesium hydroxide, with magnesium oxide or calcium carbonate and meglumine and the like.

Suitable surfactants as component can be selected from, but not limited to the group consisting of anionic surfactants, preferably sodium lauryl sulphate; polyethylene glycols (PEGs), preferably those PEGs having molecular weight in the range of about 2000 to 10000, more preferably PEG 3350, PEG 4000, PEG 6000, PEG 8000; Polysorbates, preferably Tween 20, Tween 80 or Span 80; fatty acid esters, preferably propylene glycol caprylates such as Capmul PG-8, Capryol 90; esters of glycerol and fatty acids, preferably glycerol oleates and caprylates (Capmul MCM); esters of polyethylene glycol and fatty acids, such as Labrasol and Solutol; castor oil ethoxylate (glycerol polyethylene glycol ricinoleate) such as Cremophor EL and Cremophor RH 40. More preferably the surfactant is selected from the group consisting of sodium lauryl sulphate; PEG 3350, PEG 4000, PEG 600 or, PEG 8000 and preferably PEG 6000; Tween 20 or Tween 80; and esters of polyethylene glycol and fatty acids, most preferably sodium lauryl sulphate and PEG 6000 and in particular sodium lauryl sulphate and the like.

Suitable sweeteners may be selected from, but not limited to the group consisting of aspartame, saccharin sodium, dipotassium glycyrrhizinate, aspartame, stevia, thaumatin, and the like.

Suitable film-forming agents and coating materials, if used, may include, but are not limited to hydroxypropyl methylcellulose (Hypromellose), hydroxypropyl cellulose, polyvinylalcohol, , methylcellulose, ethylcellulose, hydroxypropylmethyl cellulose phthalate, hydroxypropylmethyl cellulose acetate succinate, shellac, liquid glucose, hydroxyethyl cellulose, polyvinylpyrrolidone, copolymers of vinylpyrrolidone and vinylacetate such as Kollidon® VA64 BASF, copolymers of

acrylic and/or methacrylic acid esters with trimethylammoniummethylacrylate, copolymers of dimethylaminomethacrylic acid and neutral methacrylic acid esters, polymers of methacrylic acid or methacrylic acid esters, copolymers of acrylic acid ethylester and methacrylic acid methyl ester, and copolymers of acrylic acid and acrylic acid methylester and the like.

Suitable plasticizers, if used, may include, but are not limited to polyethylene glycol, diethyl phthalate and glycerol. Preference is given to polyethylene glycol and the like.

In an aspect the multi-particulate composition of acotiamide can be prepared by one or more of various methods, but not limited direct compression, wet granulation, dry granulation (roller compaction), solvent evaporation, hot melt granulation, hot melt extrusion, fluid bed granulation, spray drying, extrusion-spheronization and the like. Such processes are well known in the art.

In one aspect, the present specification relates to a pharmaceutical unit compositions comprising acotiamide and a proton pump inhibitor.

In one aspect, the present specification relates to a pharmaceutical unit compositions comprising multi-particulate compositions of acotiamide and a delayed release composition of a proton pump inhibitor.

In another aspect, the present invention relates to a capsule unit composition comprising:

i) plurality of multi-particulate sustained release or delayed release compositions of acotiamide and one or more rate controlling polymers, and
ii) delayed release compositions of proton pump inhibitor.

In an aspect, the present invention further relates to multi-particulate composition of acotiamide and proton pump inhibitors wherein proton pump inhibitors can be at least one selected from, but not limited to omeprazole, lansoprazole, dexlansoprazole, esomeprazole, pantoprazole, rabeprazole, ilaprazole and the like. The composition of the proton pump inhibitors are preferably in delayed release dosage forms. The delayed release dosage forms of proton pump inhibitors of the present specification releases minimal/ no release in the pH of stomach region. Such dosage form includes, are but not limited to tablets, mini-tablets, pellets, granules or other multi-particulates compositions.

In another aspect, the present invention further relates to a pharmaceutical unit composition comprising multi-particulate composition of acotiamide and a proton pump inhibitors wherein composition of acotiamide or proton pump inhibitors can be at least one or more of immediate release, sustained release or delayed release multi-particulates. In order to achieve such release profile, the said compositions can be further coated with suitable film-forming agents and/or coating materials to form a subcoating or enteric coating. Such coating methods are well known in the art.

In an aspect, the present invention relates to a capsule unit composition comprising:

i) four sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) one delayed release mini-tablet of a proton pump inhibitor.

In another aspect, the present invention relates to a capsule unit composition comprising:

i) four sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) multi-particulate delayed release compositions of a proton pump inhibitor.

In another aspect, the present invention relates to a capsule unit composition comprising:

i) three sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) multi-particulate delayed release compositions of a proton pump inhibitor.

In an aspect, the present invention relates to a capsule unit composition comprising:

i) one or more immediate release mini-tablet of acotiamide and one or more sustained release mini-tablets of acotiamide, wherein total dose of said combined mini-tablets is 300 mg acotiamide, and
ii) one delayed release mini-tablet of a proton pump inhibitor.

In an aspect, the present invention relates to a bilayer or inlay tablet unit composition comprising:

i) plurality of multi-particulate modified release compositions of acotiamide, and
ii) delayed release compositions of proton pump inhibitor.

The multi-particulate compositions of acotiamide or pharmaceutically acceptable salt thereof are preferably in solid oral dosage forms, and having sustained/delayed profile and optionally immediate release profile. Examples of such multi-particulates compositions are being selected from one or more of mini-tablets, spheroids, pellets, granules, powders, microcapsules, multiple unit particles and mixtures thereof. The multi-particulate compositions of acotiamide of the present specification can be further formulated in to a suitable dosage forms like tablets, capsules, sachets and the like. The size/dimensions of the multi-particulate composition can be modified in such a way that it can be filled in the capsule of a suitable size.

The multi-particulate modified release compositions of acotiamide further comprise one or more rate controlling polymers and one or more pharmaceutically acceptable excipients. The use of rate controlling polymer aids to achieve delayed/ sustained release profile. The modified release acotiamide compositions of the present specification are specifically designed to accommodate high drug load and also to meet the recommended total daily dose of acotiamide or pharmaceutically acceptable salts thereof in a single unit dosage form along with the desired release profile. The term drug load refers to amount of acotiamide or its pharmaceutically acceptable salt thereof present in the formulation based on the total weight of the formulation. The modified release acotiamide compositions of the present specification contains drug load at least 45% (w/w), preferably more than 50% (w/w), preferably more than 60% (w/w).

The individual components of acotiamide and proton pump inhibitor formulations of the present specification can be prepared by one or more of various methods, but not limited direct compression, wet granulation, dry granulation (roller compaction), solvent evaporation, hot melt granulation, hot melt extrusion, fluid bed granulation, spray drying, extrusion-spheronization and the like. Such processes are well known in the art.

The pharmaceutical unit composition comprising combination of acotiamide and a proton pump inhibitor can be prepared by making the individual composition or components of acotiamide or a proton pump inhibitor separately which can then be formulated in to a suitable dosage forms like tablets, capsules, sachets and the like. For example, the individual components can be filled in a capsule or a sachet. The size/dimensions of the multi-particulate composition can be modified in such a way that it can be filled in the capsule of a suitable size. Alternatively, the individual components can be further processed to form a bilayer tablet or inlay tablet.

The pharmaceutical unit compositions of the present specification may be suitably packed in to any conventional pack like stick pack, blister pack, HDPE bottle or any other packaging well known in the art.

The pharmaceutical unit compositions comprising acotiamide or acotiamide and a proton pump inhibitor of the present specification can be used for the treatment and/or management of functional dyspepsia, postprandial fullness in the functional dyspepsia, upper abdominal fullness, gastroesophageal reflux disease, and other upper gastrointestinal issues or disorders.

The pharmaceutical unit compositions of the present specification is stable throughout the shelf life which can be established by subjecting the compositions to accelerated and long term stability studies.

The following examples will further describe certain specific aspects and embodiments of the invention in greater details and are not intended to limit the scope of invention.

EXAMPLES

Example 1: Multi-particulate pharmaceutical compositions of acotiamide.

Ingredients	Example % w/w							
	1a	1b	1c	1d	1e	1f	1g	1h
Intra-granular								
Acotiamide Hydrochloride#	62.5	62.5	75.0	62.5	62.5	62.5	71.43	83.33
Hydroxy Propyl Methylcellulose (K-100 M)	6.3	-	-	-	-	-	-	-
Hydroxy Propyl Methylcellulose (K-4 M CR)	-	6.3	-	6.3	-	6.2	0.71	-
Hydroxy Propyl Methylcellulose (K-100 LV CR)	-	12.5	-	18.8	-	18.75	15.71	-
Microcrystalline Cellulose Powder (PH101)	14.6	18.1	11.0	11.9	17.5	11.8	11.43	9.16
Lactose monohydrate	14.6	-	-	-	-	-	-	6.66
Methacrylic Acid NF (Eudragit L 100-55)	-	-	10.0	-	15.0	-	-	-
Hydroxypropylmethyl Cellulose (Methocel E5)	-	-	2.0	-	3.0	-	-	-

Granulating Fluid								
Purified water	qs	qs	qs	qs	qs	qs	qs	qs
Sodium hydroxide	-	-	0.4	-	0.6	-	-	-
Extra- granular								
Talc	1.0	0.4	-	0.4	-	0.37	0.36	0.41
Magnesium Stearate	1.0	0.3	1.6	0.3	1.3	0.25	0.36	0.41
# Acotiamide HCl hydrate								

Process:

1. Acotiamide along with other intra-granular excipients were granulated by using granulating fluid in a high shear granulator.

2. Granules obtained in step 1 were then dried, milled to a suitable sized.

3. The milled granules obtained in step 2 were lubricated with extra granular material and compressed in to mini-tablets.

4. Mini tablets obtained in step 3 were filled in capsule as per required therapeutic dose.

Example 2: Multi-particulate pharmaceutical compositions of proton pump inhibitor

Rabeprazole sodium Delayed Release 20 mg		
Ingredients	Example 2a (mg)	Example 2b (mg)
Granulating material		
Rabeprazole sodium	21	21
Mannitol	75	50
Microcrystalline cellulose	-	25
Light magnesium oxide	4	4
Sodium starch glycolate	5	5
Hydroxy propyl cellulose (Klucel LF)	3	3
Blending material		
Magnesium stearate	1.0	1.0
Talc	0.4	0.4
Subcoating		
Hydroxy propyl cellulose (SL)	3.8	3.8
Ethyl cellulose (7cps)	3.2	3.2
Enteric coating		
Eudragit L100-55	16	16
Sodium hydroxide	0.3	0.3
PEG 6000	1.2	1.2
Talc	1.0	1.0
Titanium dioxide	1.0	1.0
Ferric oxide	0.10	0.10
Total weight	**136**	**136**

Process:

1. Rabeprazole along with other intra-granular excipients were granulated by using granulating fluid in a high shear granulator.

2. Granules obtained in step 1 were then dried, milled to a suitable sized.

3. The milled granules obtained in step 2 were lubricated with extra-granular material and compressed in to tablets.

4. The tablets were subcoated and followed by enteric coated.

Example 3: Multi-particulate pharmaceutical compositions of proton pump inhibitor

Ingredients	**Example 3a (mg)**	**Example 3b (mg)**
Drug coating material		
Omeprazole	20.00	40.0
Anhydrous disodium hydrogen phosphate	62.00	62.00
Mannitol 25 C	12.46	12.46
Light magnesium carbonate	1.49	1.49
Sodium lauryl sulphate	0.49	0.49
Sugar globules (#16/#18)	35.57	35.57
Hydroxypropyl methyl cellulsoe (6cps)	5.41	5.41
Drug coating solution		
Polysorbate 80	0.62	0.62
Anhydrous disodium hydrogen phosphate	0.63	0.63
Purified water	qs	qs
Outer layer material		
Hydroxy propyl methyl cellulose (6cps)	0.77	0.77
Mannnitol 25C	1.39	1.39
Purified water	qs	qs
Sub coating material		
Omeprazole drug coated pellets	140.83	160.83
Hydroxy propyl methyl cellulose (6cps)	9.16	9.16
Purified water	qs	qs
Enteric coating material		
Omeprazole sub coated pellets	150.00	170.00
Methacrylic acid coploymer (Type C) Eudragit L 100 (55)	34.99	34.99
Sodium hydroxide pellets	0.47	0.47
Polyethylene glycol	4.20	4.20
Titanium dioxide	1.84	1.84
Talc	3.50	3.50
Purified water	qs	qs
Total weight	**195.0**	**215.0**

Process:

1. Drug coating solution was prepared by dissolving dibasic sodium phosphate and Poysorbate 80 until it dissolved.
2. Then HPMC, Omeprazole Light magnesium carbonate, SLS, Mannitol were added in the drug coating solution and mixed well.
3. The mixture obtained in Step 2 was coated on sugar spheres in Fluidized Bed Processor.
4. The pellets obtained in step 3 were subcoated and followed by enteric coated.

Example 4: Multi-particulate pharmaceutical compositions of acotiamide and proton pump inhibitor

Some multi-particulate pharmaceutical combinations of acotiamide and a proton pump inhibitor as per the present specification are described below:

Examples	Drug*	Dosage form	Dose
Example 4a	Acotiamide	Sustained Release multi-particulate	300 mg
	Omeprazole	Delayed release pellets	20 mg/10 mg
Example 4b	Acotiamide	Sustained Release multi-particulate	300 mg
	Rabeprazole	Delayed release mini-tablet	20 mg
Example 4c	Acotiamide	Immediate release and sustained Release mini-tablets	300 mg
	Rabeprazole	Delayed release mini-tablet	20 mg
Example 4d	Acotiamide	Sustained Release multi-particulate	300 mg
	Pantoprazole	Delayed release mini-tablet	20 mg/40 mg
Example 4e	Acotiamide	Sustained Release multi-particulate	300 mg
	Lansoprazole	Delayed release pellets	20 mg
Example 4f	Acotiamide	Sustained Release multi-particulate	300 mg
	Esomeprazole	Delayed release pellets	20/40 mg
Example 4g	Acotiamide	Sustained Release multi-particulate	300 mg
	Dexlansoprazole	Delayed release pellets	30 mg/60 mg
Example 4h	Acotiamide	sustained Release mini-tablets	300 mg
	Rabeprazole	Delayed release pellets	20 mg
Example 4i	Acotiamide	Immediate release and sustained release mini-tablets	300 mg

	Rabeprazole	Delayed release pellets	20 mg
*Drug or pharmaceutically acceptable salts thereof			

Process:

1. Multi-particulate pharmaceutical compositions of acotiamide were prepared according to example 1.
2. Multi-particulate pharmaceutical compositions of a proton pump inhibitor were prepared according to example 2 or 3.
3. The compositions prepared in Step 1 & 2 were filled in a capsule.

Example 5

Stability Study:

The stability of the pharmaceutical unit composition comprising multi-particulate modified release compositions of acotiamide as per the present specification was evaluated through accelerated stability studies. The composition prepared according to the formula and process of example 1f was subjected to stability study at 40 °C/75% RH. The compositions was found to be stable at accelerated conditions. Table 1 represents the study result data.

Table 1: Stability study of example 1f

	Example- 1f			
Stability condition	40 °C/75% RH			
Time period	Limit	Initial	1M	3M
Assay (%)	90-110	100.50	99.10	101.53
Related Substance (%)	NMT 2	0.15	0.29	0.16

Example 6

In-vitro dissolution study:

The release profile of the pharmaceutical unit composition comprising multi-particulates modified release compositions of acotiamide as per the present specification was evaluated through *in vitro* dissolution studies. The unit composition was prepared according to the formula and process of example 1f, and was subjected to an *in vitro* dissolution study in a changeover dissolution medium at 100 rpm, 37°C by using USP type I apparatus. Table 2 represents the dissolution results data. Figure 1 represent the dissolution profile of acotiamide hydrochloride composition of example 1f at initial stage. The dissolution profile of the composition was also assessed after storage in accelerated stability condition (40 °C/75% RH) for 1 month and 3 months, the data is represented in table 2.

Table 2: Dissolution study of Example 1f

		Acotiamide HCl hydrate release profile			
Time point	**Time (hour)**	**Limit (% of release)**	**Initial (% of release)**	**1M (% of release)**	**3M (% of release)**
Dissolution study of example 1f		900 mL 0.01N HCl, , 100RPM			
	1	NMT 55	12	12	7
		900mL pH 6.8 Phosphate buffer saline + 0.5% SLS, 100RPM			
	3	At least about 20-75	34	32	22
	9	NLT 50	69	69	54

NMT - Not more than; NLT - Not less than

Example 7:

The release profile of the pharmaceutical unit composition comprising multi-particulate modified release compositions of acotiamide and delayed release compositions of proton pump inhibitors as per the present specification was evaluated through *in vitro* dissolution studies. The unit composition was prepared according to

the formula and process of example 4h, and was subjected to an *in vitro* dissolution study in a changeover dissolution medium at 100 rpm, 37°C by using USP type I apparatus. Table 3 represents the dissolution results data of acotiamide or pharmaceutically acceptable salts thereof in example 4h. Table 4 represents the dissolution results data of rabeprazole or pharmaceutically acceptable salts thereof.

Table 3: Dissolution study of acotiamide in example 4h

		Acotiamide HCl Hydrate release profile	
Time point	**Time (hour)**	**Limit (% release)**	**(% of release)**
Dissolution study of example 4h	900 mL 0.01N HCl, , 100 rpm		
	1.0	NMT 55%	13
	900 mL pH 6.8 Phosphate buffer saline+ 0.5% SLS, 100 rpm		
	3	At least about 20 % – 75%	29
	9	NLT 50%	59

Table 4: Dissolution study of rabeprazole in example 4h

		Rabeprazole release profile
Time point	Time (hour)	(% of release)
Dissolution study of example 4h	900 mL 0.01N HCl, , 100 rpm	
	0.0	0
	1.0	0
	2.0	0
	900 mL pH 8 Phosphate buffer + 0.5% SLS, 100 rpm	
	10 min	8
	20 min	86
	30 min	95

We Claim:

1. A pharmaceutical unit composition comprising:
   i) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,
   ii) optionally multi-particulate immediate release compositions of acotiamide or pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily.

2. The composition of claim 1, wherein the total dose of acotiamide or pharmaceutically acceptable salts thereof in the unit composition is about 300 mg.

3. The composition of claim 1, wherein acotiamide or pharmaceutically acceptable salts thereof is present in an amount of at least 50% by weight of the total unit composition.

4. The composition of claim 1, wherein the unit composition is selected from capsule, tablet and sachet.

5. The composition of claim 1, wherein the unit composition is capsule.

6. The composition of claim 1, wherein the multi-particulate compositions of acotiamide or pharmaceutically acceptable salts thereof are selected from one or more of mini-tablets, spheroids, pellets, granules, powders, microcapsules, multiple unit particles and mixtures thereof.

7. The composition of claim 1, when subjected to *in vitro* changeover dissolution study exhibits following release profile:
   i) releases not more than 55% in 1 hour, in 900 ml of 0.01N HCl medium at 100 rpm and 37°C using USP type I apparatus, and
   ii) releases at least about 20% - 75% in 3 hours and not less than 50% in 9 hours, in 900 ml of pH 6.8 phosphate buffer medium saline with 0.5% Sodium laureth sulfate at 100 rpm and 37°C using USP type I apparatus.

8. A pharmaceutical unit composition comprising:

i) delayed release compositions of a proton pump inhibitor or pharmaceutically acceptable salts thereof,

ii) plurality of multi-particulate modified release compositions of acotiamide or pharmaceutically acceptable salts thereof and one or more rate controlling polymers,

iii) optionally multi-particulate immediate release compositions of acotiamide or a pharmaceutically acceptable salts thereof,

wherein the said unit composition is administered once daily.

9. The composition of claim 8, wherein the proton pump inhibitor is selected from rabeprazole, omeprazole, esomeprazole, ilaprazole, lansoprazole, pantoprazole, leminoprazole or an optically active isomer thereof, or pharmaceutically acceptable salts thereof.

10. The composition of claim 8, wherein the proton pump inhibitor is selected from rabeprazole, omeprazole or an optically active isomer thereof, or pharmaceutically acceptable salts thereof.

11. The composition of claim 8, wherein the delayed release compositions of a proton pump inhibitor are selected from one or more of tablet, mini-tablets or pellets or granules.

12. The composition of claim 8, wherein the total dose of acotiamide or pharmaceutically acceptable salts thereof in the unit composition is 300 mg.

13. The composition of claim 8, wherein the acotiamide or pharmaceutically acceptable salts thereof is present in an amount of at least 50% by weight of the total multi-particulate acotiamide compositions.

14. The composition of claim 8, wherein the modified release compositions of acotiamide are selected from sustained release, delayed release or mixtures thereof.

15. The composition as claimed in claim 8, wherein the multi-particulate compositions of acotiamide are selected from one or more of mini-tablets, spheroids, pellets, granules, powders, microcapsules, multiple unit particles and mixtures thereof.

16. The composition of claim 8, wherein the said acotiamide and proton pump inhibitor compositions are physically separated.

17. The composition of claim 8, wherein the unit composition is selected from capsule, tablet and sachet.

18. The composition of claim 8, wherein the unit composition is capsule.

19. The use of composition as claimed in claim 8, for the treatment and/or management of functional dyspepsia, postprandial fullness in the functional dyspepsia, upper abdominal fullness, gastroesophageal reflux disease, and other upper gastrointestinal issues or disorders.

20. A pharmaceutical unit composition of proton pump inhibitor and acotiamide or a pharmaceutically acceptable salts thereof, wherein the composition is prepared by the steps comprising:

i) preparing a delayed release compositions of a proton pump inhibitor,
ii) preparing one more modified release mini-tablet compositions of acotiamide and optionally preparing immediate release mini-tablet compositions of acotiamide,
iii) filling the compositions of step (i) and step (ii) in a capsule.

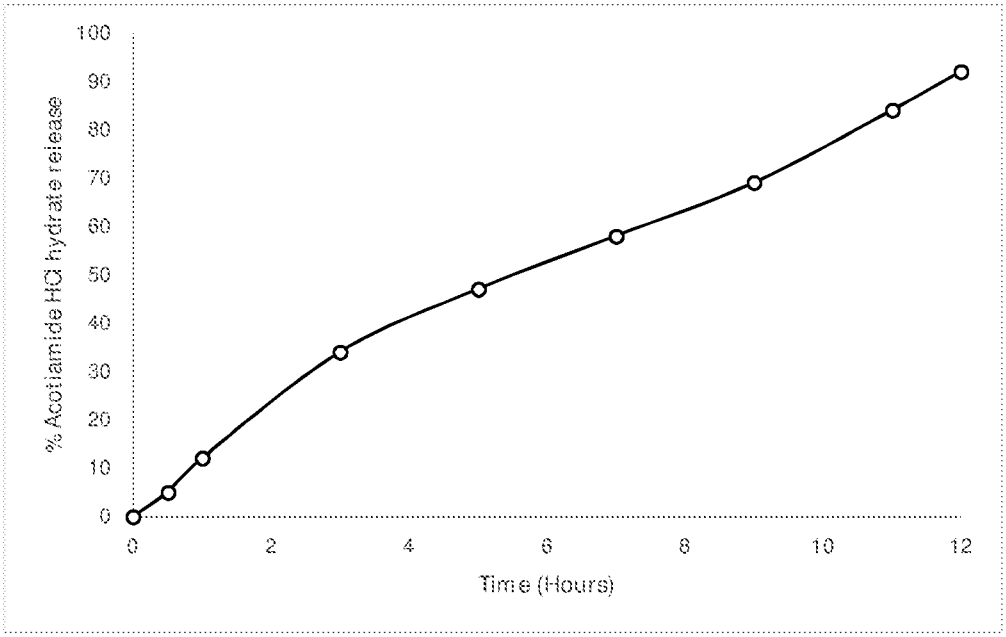
% Acotiamide HCl hydrate release
100
90
80
70
60
50
40
30
20
10
0
0
2
4
6
8
10
12
Time (Hours)

Figure: 1

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2019/059957

A. CLASSIFICATION OF SUBJECT MATTER
A61K31/295 Version=2020.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

TotalPatent One, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	CN106511291A (JINAN KANGHE MEDICAL TECH CO LTD, 22 MARCH, 2017) the whole document particularly Abstract	1-7
Y	IN201721028736A (SOLUTIONS IN INNOVATIVE RESEARCH PVT. LTD., 15 SEPTEMBER, 2017) the whole document	8-20
Y	Rakesh N. Tirpude et al "Rabeprazole sodium delayed-release multiparticulates: Effect of enteric coating layers on product performance", Journal of Advanced Pharmaceutical Technology & Research, Jul-Sep 2011; vol. 2(3), 184-191. the whole document	8-20
Y	Volnei Jose Tondo Filho, et al "Development of a multiparticulate system containing entericrelease mini-tablets of omeprazole", Brazilian Journal of Pharmaceutical Sciences vol. 50, no. 3, july/sep., 2014, http://dx.doi.org/10.1590/S1984-82502014000300008 the whole document	8-20

☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
27-02-2020	27-02-2020
Name and mailing address of the ISA/ Indian Patent Office Plot No.32, Sector 14,Dwarka,New Delhi-110075 Facsimile No.	Authorized officer Ravi S Telephone No. +91-1125300200

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2019/059957

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	NS Dey et al. "Multiparticulate Drug Delivery Systems for Controlled Release", Tropical Journal of Pharmaceutical Research, September 2008; 7 (3): 1067-1075. the whole document	1-20
A	WO2000025752A1 (John G. Devane et al., 11 MAY, 2000) the whole document	1-20
A	WO2005032513A2 (Sripriya Venkata Ramana Rao et al., 14 APRIL, 2005) the whole document	1-20
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A	US20060240105A1 (RECRO GAINESVILLE LLC, 26 OCTOBER, 2006) the whole document	8-20
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